as a fundamental notion with “what the clock reads in this part of the overall quantum superposition.” In that way, time has emerged from a static state, thanks to the magic of entanglement.

The jury remains out on whether the energy of the universe actually is zero, and therefore time is emergent, or it is any other number, such that time is fundamental. At the current state of the art, it makes sense to keep our options open and investigate both possibilities.

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* In 2013, Juan Maldacena and Leonard Susskind suggested that we should think of entangled particles as being connected by a microscopic (and impossible-to-travel-through) wormhole in spacetime. This has been dubbed the “ER=EPR conjecture,” after two famous papers from 1935: one by Einstein and Nathan Rosen, where they introduced the concept of wormholes; and the other of course by Einstein, Rosen, and Boris Podolsky, where they discussed entanglement. How far such a suggestion can be taken is still unclear.